

MATH 164 Discussion (Week 1): Analysis Prereqs for Optimization (TA Key)

Answer key version.

A. Limit points, closed sets, compact sets

Limit point (accumulation point). A point x is a limit point of $S \subset \mathbb{R}^n$ if every open ball $B(x, r)$ contains a point of S different from x .

Closed set. S is closed if it contains all its limit points (equivalently: $x_k \in S$ and $x_k \rightarrow x$ implies $x \in S$).

Compact (in $\mathbb{R}^n$). S is compact iff it is **closed and bounded** (Heine–Borel).

Optimization link. Compact feasible set + continuous objective $\Rightarrow$ minimizer exists.

Quick checks: sets and limit points (Answers) ✓

1. Which sets are closed?

$(0, 1) \subset \mathbb{R}$ (closed? **no**), $[0, 1]$ (**yes**), $\{1/n : n \in \mathbb{N}\}$ (**no in $\mathbb{R}$**).

Note: $\{1/n\}$ does not contain its limit point 0 (in $\mathbb{R}$), hence not closed.

2. For $S = (0, 1)$: list two limit points of S in $\mathbb{R}$.

0 and 1 (indeed every neighborhood of 0 or 1 meets $(0, 1)$).

3. True/False: If S is closed, then it is bounded. **False** (e.g. $\mathbb{R}$ is closed but unbounded).

Optimization example: existence can fail (Answers)

One standard example:

$S=(0,1)$, $f(x)=x$.

Then $\inf_{x \in (0,1)} x = 0$ but $0 \notin S$, so no minimizer exists on S . (Alternative: $S = \mathbb{R}$, $f(x) = e^{-x}$ has $\inf = 0$ not attained.)

B. Continuity and why we care

Continuity at x . f is continuous at x if $x_k \rightarrow x \Rightarrow f(x_k) \rightarrow f(x)$.

Extreme Value Theorem (EVT). If $S \subset \mathbb{R}^n$ is compact and $f : S \rightarrow \mathbb{R}$ is continuous, then f attains both a minimum and a maximum on S .

Optimization link. Existence of solutions to constrained problems often reduces to verifying:

(i) feasibility set compact + (ii) objective continuous.

EVT for Existence of Extremizer (Answers) ✓

1. $\min_{x \in [0,1]} (x - 0.3)^2$ Guaranteed? **Yes**. Reason: $[0, 1]$ **compact**; f **continuous**.

2. $\min_{x \in (0,1)} (x - 0.3)^2$ Guaranteed? **Not by EVT (set not compact)**. *Remark:* Here the minimizer **does** exist at $x^* = 0.3$ since $0.3 \in (0, 1)$, but EVT cannot be invoked.

3. $\min_{x \in \mathbb{R}} e^{-x^2}$ Guaranteed? **No**. Reason: $\mathbb{R}$ **not compact**; $\inf = 0$ **not attained**.

B*. Linear algebra basics (Answers)

1. (T/F) Multiplying a column of A by a nonzero scalar does not change $\text{rank}(A)$. **T**
2. (T/F) Interchanging two columns could change $\text{rank}(A)$. **F**
3. (T/F) Adding to a column a linear combination of other columns does not change $\text{rank}(A)$. **T**
4. (T/F) $Ax = b$ has a solution iff $\text{rank}(A) = \text{rank}([A \mid b])$. **T**
5. If $A \in \mathbb{R}^{m \times n}$ and $\text{rank}(A) = m$, then the number of free variables (when solvable) is $n - m$.
6. (T/F) All eigenvalues of a real symmetric matrix are real. **T**
7. (T/F) Any real symmetric $n \times n$ matrix has an orthogonal eigenbasis. **T**
8. Fill in:

$$\mathcal{R}(A)^\perp = \mathcal{N}(A^\top), \quad \mathcal{N}(A)^\perp = \mathcal{R}(A^\top).$$
9. (T/F) P is an orthogonal projector iff $P^2 = P = P^\top$. **T**
10. (T/F) For symmetric Q , $Q \succ 0$ iff all eigenvalues of Q are positive. **T**
11. Frobenius norm:

$$\left\| \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix} \right\|_F = \sqrt{1+4+4+0} = \sqrt{9} = 3\sqrt{1+4+4+0} = \sqrt{9} = 3$$

Short application (optional, Answers).

- $P_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$: **Yes**, since $P_1^2 = P_1$ and $P_1^\top = P_1$.
- $P_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$: **No**, since $P_2^\top \neq P_2$ (so it is not an *orthogonal* projector).

C. Derivatives, gradients, and first-order conditions

1D derivative. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

Gradient (multivariable). For $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla f(x)$ satisfies

$$f(x+d) \approx f(x) + \nabla f(x)^\top d \quad (\text{small } d).$$

First-order necessary condition (unconstrained). If f is differentiable and x^* is a local minimizer in the interior of the domain, then

$$\nabla f(x^*) = 0.$$

(Second-order: $\nabla^2 f(x^*) \succeq 0$ is typical for local minima.)

Optimization link: 1D search shows up via line search. Given direction d , define $\phi(\alpha) = f(x + \alpha d)$ and minimize over $\alpha > 0$.

Compute + interpret (Answers)



Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q = Q^\top$.

1.

$$\nabla f(x) = Qx - b, \quad \nabla^2 f(x) = Q.$$
2. If Q is positive definite, then f has a **unique** minimizer. Why?
Because $Q \succ 0$ implies f is strictly convex (Hessian $\succ 0$), so the stationary point is unique global minimizer.
3. What if
 - (a) Q is Positive Semi-Definite while not Positive Definite?
 f is convex but not strictly convex. Minimizers may be non-unique (a flat direction). Also a minimizer exists iff $b \in \mathcal{R}(Q)$; otherwise f is unbounded below along $\mathcal{N}(Q)$.
 - (b) Q has negative eigenvalues?
 f is non-convex and typically unbounded below (since $x^\top Qx$ can go to $-\infty$ along a negative-eigenvalue direction), hence no minimizer in general.
 - (c) Q is not symmetric?
The quadratic form only depends on the symmetric part $Q_s = \frac{1}{2}(Q + Q^\top)$. Then $\nabla f(x) = Q_s x - b$, $\nabla^2 f(x) = Q_s$. A unique minimizer exists iff $Q_s \succ 0$; the optimality condition is $Q_s x^* = b$.

4. Solve $\nabla f(x) = 0$: the minimizer satisfies

$$Qx^* = \mathbf{b} \quad \Rightarrow \quad x^* = \mathbf{Q}^{-1}b\mathbf{Q}^{-1}b\mathbf{Q}^{-1}b\mathbf{Q}^{-1}b(\text{if } Q \text{ invertible; in PD case it is}).$$

Line-search chain rule (Answers)

Let $\phi(\alpha) = f(x + \alpha d)$. Then

$$\phi'(\alpha) = \nabla f(\mathbf{x} + \alpha \mathbf{d})^\top d \nabla f(\mathbf{x} + \alpha \mathbf{d})^\top d \nabla f(\mathbf{x} + \alpha \mathbf{d})^\top d \nabla f(\mathbf{x} + \alpha \mathbf{d})^\top d.$$

A “best step size” often solves $\phi'(\alpha) = 0$. One 1D method: **bisection** / **Newton** / **secant** / **golden-section** (any one acceptable).